tm1514 recheck at k = 24: results

Protocol `XP-20261106-tm1514-recheck`, `PROTOCOL.json` version 2 and `PROTOCOL-AMENDMENT-1.md` (counted-only; C3 deferred). 2026-11-10.

Standing. Counted operations only, from two local `ecbench` sessions on a development host at isolation L0 (one shared 4-vCPU KVM guest, Intel Xeon 2.10 GHz). No wall time, cycle count or isolation level enters any result. The study has no `VALIDATION.json`, no independent replay certificate and no validator report under the validation gate (`docs/VALIDATION_GATE.md`), so it is not an admitted benchmark or leaderboard row. Every number below is copied from the frozen analysis output and is recorded as statement claim `PCC-20261110-97c3b6`, standing unverified [pcc:unverified].

Result

On `icv1-fp24-tm1514-dce9313c`, rho with negation (`rho-neg`) is cheaper than the ledger incumbent (`inc`, `wwalkGg-x09-sr-n2`) at every weight of the §5a band [pcc:unverified]. The inc/rho-neg ratio of mean S, over 1,152 paired solves per arm, is:

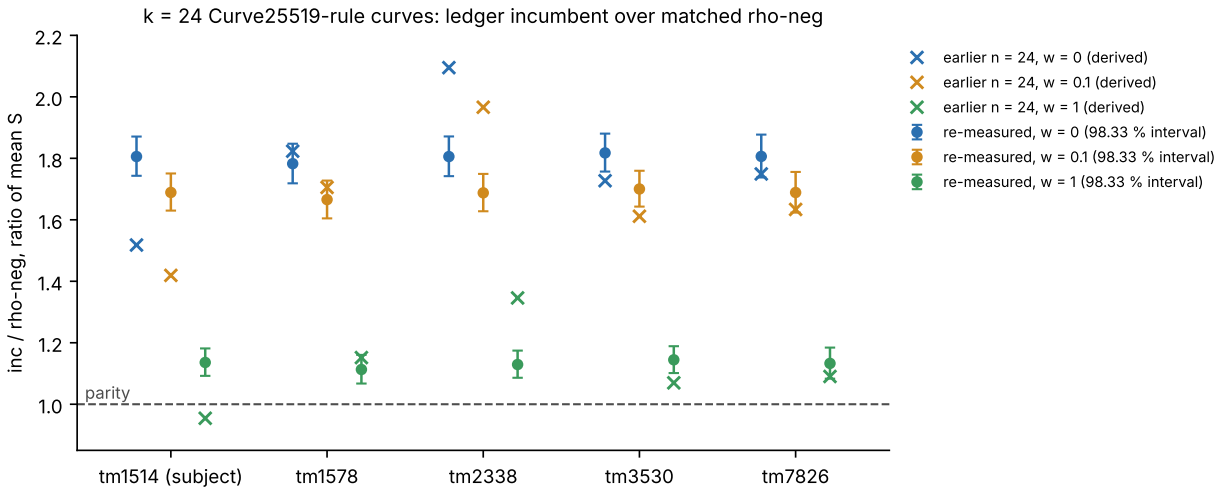
w (GAE per native op)	R_tm1514 (95 % interval)	C4 verdict
0	1.8058 [1.7543, 1.8598]	R_tm1514 above 1
0.1	1.6890 [1.6402, 1.7404]	R_tm1514 above 1
1	1.1359 [1.1001, 1.1739]	R_tm1514 above 1

tm1514 sits within its four siblings' ratio. The earlier 1.52 at w = 0 (and 0.95 at w = 1), from 24 solves per arm in the 4 November multi-curve session, was a small-sample draw. [pcc:unverified]

claim	frozen rule	outcome
C0	every row verified; twins reproduce partners exactly	holds (no failures, nothing censored)
C1, w = 0	98.33 % interval of D = ln R_tm1514 − ln R_siblings, against the A/A half-width	D 0.0016 [−0.0378, 0.0416], A/A half-width 0.0670: within the sibling ratio (draw not excluded)
C1, w = 0.1	as above	D 0.0021 [−0.0379, 0.0427], A/A half-width 0.0680: within the sibling ratio (draw not excluded)
C1, w = 1	as above	D 0.0053 [−0.0389, 0.0501], A/A half-width 0.0743: within the sibling ratio (draw not excluded)
C1 (secondary, non-overlap)	tm1514's R interval entirely below the siblings'	within the sibling ratio at all three weights
C1b	95 % interval of R_tm1514(0) excludes 1.52	1.8058 [1.7543, 1.8598]: the earlier 1.52 was a draw
C2 (positive control)	pooled rho-plain/rho-neg 95 % lower end ≥ 1.20, every curve > 1.10	1.2821 [1.2609, 1.3030]; per curve 1.2549 to 1.3076: detected
C3	measured native price	deferred (AMENDMENT-1 §2.4)
C4	95 % interval of R_tm1514(w) against 1	above 1 at w = 0, 0.1 and 1

The pooled sibling ratio R_siblings is 1.8029 [1.7717, 1.8358] at w = 0, 1.6855 [1.6560, 1.7168] at w = 0.1 and 1.1298 [1.1074, 1.1535] at w = 1 (98.33 %). Per curve (point values; 98.33 % intervals in `result.json` and Figure 1):

curve	role	R(0)	R(0.1)	R(1)	rho-plain/rho-neg (w = 0)
icv1-fp24-tm1514-dce9313c	subject	1.8058	1.6890	1.1359	1.2993
icv1-fp24-tm1578-e83ed1b2	sibling	1.7826	1.6657	1.1131	1.2619
icv1-fp24-tm2338-191c1070	sibling	1.8057	1.6877	1.1292	1.2549
icv1-fp24-tm3530-7cdfa218	sibling	1.8176	1.7002	1.1446	1.2874
icv1-fp24-tm7826-eab5872b	sibling	1.8063	1.6888	1.1329	1.3076



1,152 inc/rho-neg pairs per curve (96 targets x 6 rounds x 2 replicates). Sessions A3 ECBS1h250938713613, B3 ECBS1hdd033a38d490. Counted operations; native work priced at w GAE/op. x: earlier 4 Nov session (24 solves per arm), inc S / rho-neg S from band-k24.txt. Above 1: rho-neg cheaper. Source: analysis/result.json.

Figure 1: Per-curve inc/rho-neg ratio at w = 0, 0.1 and 1 with 98.33 % intervals; crosses are the earlier n = 24 values (inc S / rho-neg S from band-k24.txt, derived). Dashed: parity.

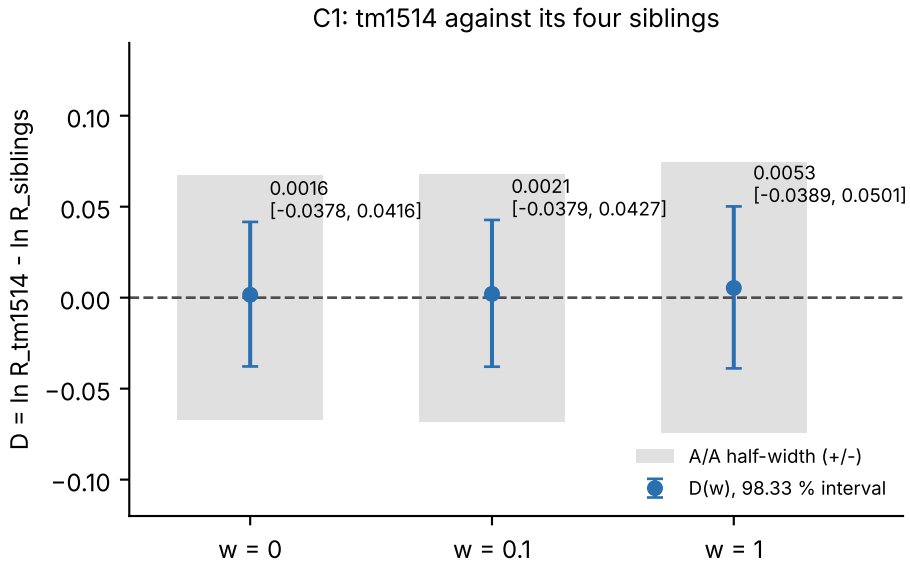


Figure 2: D(w) = ln R_tm1514 − ln R_siblings with 98.33 % intervals; grey band: the seed-independent A/A half-width at each w.

The earlier 24-solve values (derived from `band-k24.txt`) fall on both sides of the new 98.33 % intervals: outside them at all three weights on tm1514 (below), tm2338 (above) and tm3530 (below), and inside them on tm1578 and tm7826.

Design and data

- Question: is tm1514's 1.52 a draw, or does it sit below its four family siblings (`PROTOCOL.md`)? Five `prime_explicit` curves of `research/curve25519_autolab_20261104/multicurve/spec-k24.json`, p near 2^24, r ≈ 2^21; arms `rho-neg`, `rho-neg-aa`, `inc`, `inc-aa`, `rho-plain`; 96 targets per curve from `target_seed` 546787239895877; 6 measured rounds and 1 warmup.
- Protocol: `PROTOCOL.json` version 2, protocol_sha256 `d348123a...81d6`, frozen 2026-11-10T17:17:31Z before any A3/B3 record existed; `PROTOCOL-AMENDMENT-1.md` governs where they differ.
- Sessions (spec id, seed): A3 `ECBS1h250938713613` (spec `ECS1h3d4c0d4813f4`, seed 8671006997891351) and B3 `ECBS1hdd033a38d490` (spec `ECS1h58d3eafbe575`, seed 6232168080843969). 16,800 executions each, all `verified` (2,880 per arm). One locally built release `ecbench`, sha256 `b8570e7f...5603` in each `host.json`. [pcc:unverified]
- Audit: `ecbench verify --replay-all` on each session: `ok`, 16,800 records, 16,800 verified, 14,400 replays reproduced, 0 problems (`audit/verify-A3.json`, `audit/verify-B3.json`). These are self-audits by the same binary on the same host, not an independent replay certificate.
- Analysis: `examples/tm1514_analyze.rs` (sha256 `6534cfe1...f197`), run as `tm1514_analyze --a sessions/A3 --b sessions/B3 --out analysis`. Arm labels were masked while the verdicts were computed (key sha256 `4927ed61...ef85`). Paired percentile bootstrap over targets, stratified by curve, B = 10,000, seed 244596609222896; 98.33 % (Bonferroni over three weights) for C1 and the per-curve band ratios, 95 % otherwise. Output `analysis/result.json` (sha256 `c7c473de...d236`) and `analysis/summary.txt` (`1b9bb451...a596`).
- Raw data are in the artifact store (AGENTS.md §13): bucket `setec-astronomy-artifacts`, key `store/sha256/<aa>/<hex>`. The manifests (`.artifacts.json`) in `sessions/A3`, `sessions/B3` and `audit` name them; `artifacts.py pull` (in `tools/artifacts/`) on this directory hydrates them. The benchmark record is `benchmarks/records/20261110T1729Z-tm1514-recheck-a3b3.json`.

file	bytes	sha256
sessions/A3/records.jsonl	127,081,439	ad6ba575...0423
sessions/A3/plan.json	3,198,831	fd35135d...4b2e
sessions/B3/records.jsonl	127,104,720	0a03a9fc...190e
sessions/B3/plan.json	3,198,856	1b98ca5c...5710
audit/verify-A3.json	1,822,675	74d0abc8...f2e7
audit/verify-B3.json	1,822,675	bb77b988...bee4

- Figures: `figures/make_figures.py` reads `analysis/result.json` (and checks its sha256) and, for the earlier markers only, `research/curve25519_autolab_20261104/multicurve/band-k24.txt`. It computes no statistic; the earlier ratios are inc S / rho-neg S from that table and are labelled derived. `figures/build_report.sh` builds `REPORT.pdf`.

What did not count

- **Replicate A is a dry run.** The EKS submission (GitHub run 42269476702) ended with exit 23 and no pod; the inferred cause is the L2 gate's 1 ms floor on per-execution solve time. Its records were never read and its seeds are not reused (AMENDMENT-1 §2.1).
- **Replicates A2 and B2 are withdrawn.** Development checks ran reduced copies of their specs on their real seeds and targets, so those seeds and targets were dropped before any full session; A3 and B3 use a new `target_seed` (AMENDMENT-1 §2.5).
- **C3 is deferred.** The price of one native operation in group additions needs timed samples of at least 5 ms each, which this instrument cannot give at k = 24; it goes to a separate protocol. The band weights 0, 0.1 and 1 are the only prices used here.
- Two revisions of the executor card fixed runner-script errors without changing data or analysis: the benchmark-record path was read from the last line of `benchrec-path.txt` (revision 1), and the verify stop condition read `ecbench verify`'s summary from stderr instead of stdout (revision 2).

Scope

The result concerns these five k = 24 Curve25519-rule miniatures, this incumbent and this instrument. Four siblings cannot rank tm1514 as unusual (best attainable rank p = 1/5); C1 asks only whether tm1514 differs from their pooled ratio, and it does not detectably. Nothing here is a statement about the 255-bit curve. The w = 0 ratios near 1.8 are close to the F = 2 collision floor of 1.875 derived in `research/curve25519_rho_parity_bound_20261104/`.